

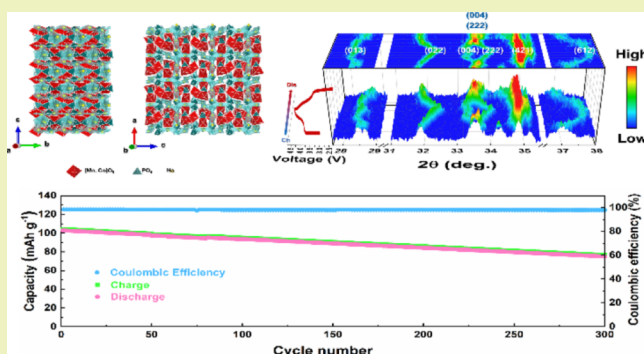
Development of a New Mixed-Polyanion Cathode with Superior Electrochemical Performances for Na-Ion Batteries

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Supporting Information

ABSTRACT: We report on $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a high-energy-density cathode material for sodium-ion batteries. Detailed structural information for $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ is obtained from Rietveld refinement of X-ray diffraction data. We confirm that 2.7 mol Na^+ ions per formula unit (116 mA h g^{-1}) can be (de)intercalated from the host $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ structure using first-principles calculation and various electrochemical tests. Moreover, $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ exhibits that 73% of the capacity at C/10 is retained even at 10 C (84 mA h g^{-1}). Notably, $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ delivers a high capacity retention of $\sim 75\%$ compared with the initial capacity for 300 cycles with a high Coulombic efficiency of more than 99%. The redox reaction mechanism of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ during charge/discharge is also confirmed through combined studies using first-principles calculation and operando X-ray absorption near-edge structure analyses.

KEYWORDS: mixed-polyanion, inductive effect, first-principles calculation, cathode, sodium, battery



INTRODUCTION

Environmental concerns have led to the increased adoption of eco-friendly, efficient, and sustainable energy storage systems.^{1–6} Li-ion batteries (LIBs) have been widely used as portable power sources for electronic devices, with their use recently expanding to vehicle and grid-scale applications owing to their high energy density and long cycle life.^{7–9} However, the exponentially growing demand for LIBs necessitates the development of LIB alternatives as Li sources are limited and their global distribution is uneven.^{10,11} To be competitive with LIBs, these alternative systems must possess high energy densities while also being safe and low cost. Na-ion batteries (NIBs) have recently received particular attention because of the readily accessible and essentially unlimited Na sources in the sea and, more importantly, the similar chemistries of NIBs and LIBs.^{12–16} Although many novel electrode materials for NIBs have been investigated, one major obstacle is the large ionic radius of Na^+ (1.02 Å) relative to that of Li^+ (0.76 Å), which results in continuous phase transitions that induce large volume changes in the cathode. Structural degradation resulting from this volume change is more apparent in O3, P2, and P3 layered compounds than in three-dimensional compounds such as polyanion- and Prussian-blue-based compounds.^{17–24} Another obstacle is the low operation voltage of NIB cathode

materials, which results from the high standard hydrogen electrode (SHE) potential of Na (-2.7 V vs SHE) relative to that of Li (-3.05 V vs SHE). To compensate for this low operation voltage and prepare high-energy-density NIBs, high-voltage cathode materials need to be developed.

As Goodenough et al. reported the inductive effect derived from phosphate (PO_4), whereby the high electronegativity of phosphor can lead to an increase in the redox potential, numerous investigations have been directed toward phosphate-based electrode materials for LIBs and NIBs.^{25–28} For NIBs, earlier works focused on Mn-based polyanions such as $\text{Na}_2\text{MnP}_2\text{O}_7$, $\text{Na}_2\text{MnPO}_4\text{F}$, and $\text{Na}_3\text{MnPO}_4\text{CO}$ owing to the high redox potentials based on the $\text{Mn}^{2+}/\text{Mn}^{3+}$ redox reaction supported by the inductive effect with high electronegativity.^{29–33} Notably, we reported a redox potential of approximately 3.8 V versus Na^+/Na for $\text{Na}_4\text{Mn}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, which is higher than that of any other Mn-based polyanion material for NIBs.³⁴ However, the Mn^{3+} ion in MnO_6 octahedra leads to cooperative Jahn–Teller distortion, which must be diminished to achieve improved capacity retention.^{35–38} To address this issue,

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many studies have focused on stabilizing the crystal structure of Mn-based electrode materials by substituting Mn ions with other metal ions.³⁹ Inspired by LiCoPO_4 , which is activated by the $\text{Co}^{2+}/^{3+}$ redox reaction, Nose et al. proposed $\text{Co}^{2+}/^{3+}$ redox in $\text{Na}_4\text{Co}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, which has an average operation voltage of 4.5 V versus Na^+/Na with good capacity retention and rate capability; however, the decomposition of the carbonate-based electrolyte at high voltage remains to be resolved.^{40–44} $\text{Na}_4\text{Mn}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ and $\text{Na}_4\text{Co}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ both exhibit stepwise voltage plateaus, which indicate the occurrence of a two-phase reaction that induces a large volume change upon de/sodiation.

In an attempt to reduce the Mn^{3+} concentration in the $\text{Mn}^{2+}/^{3+}$ redox and boost the operation voltage of $\text{Na}_4\text{Mn}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, we herein introduce Co-substituted $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a high-energy-density cathode material for NIBs.^{45–47} Through combined studies using first-principles calculation and various experiments, it is verified that $\text{Mn}^{2+}/^{3+}$ and $\text{Co}^{2+}/^{3+}$ redox pairs stimulate the activity of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ with two smooth voltage plateaus, delivering a large discharge capacity of approximately 116 mA h g^{-1} at C/10 ($1 \text{ C} = \sim 128 \text{ mA h g}^{-1}$), which corresponds to the insertion of 2.7 mol Na^+ ions per formula unit, with an average operation voltage of approximately 4.0 V versus Na^+/Na . The $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ electrode is active even at 10 C (84 mA h g^{-1}), delivering 73% of the capacity measured at C/10. It is notable that the related $\text{Mn}^{2+}/^{3+}$ and $\text{Co}^{2+}/^{3+}$ redox reaction progresses via a one-phase reaction, as confirmed by operando X-ray diffraction (XRD), where the volume change is sufficiently small (5.7%) compared with that reported in earlier work for $\text{Na}_4\text{Mn}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (7%). It is probable that the cycling stability for 300 cycles at 2 C, with capacity retention of approximately 75%, may be attributed to this single-phase reaction. A combination of theoretical and experimental investigation suggests the long-term electrochemical activity of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a promising cathode material for NIBs.

■ EXPERIMENTAL SECTION

Synthesis of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$. $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ powder was synthesized via a solid-state reaction. $\text{Na}_4\text{P}_2\text{O}_7$ (Duksan, 98.5%), $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (Alfa Aesar, 93%), CoC_2O_4 (Alfa Aesar, 98%), $\text{NH}_4\text{H}_2\text{PO}_4$ (Daejung, 98%), and 10 wt % pyromellitic acid (PA, $\text{C}_6\text{H}_4\text{O}_2$) (Alfa Aesar, 96%) in a molar ratio of 1:1.5:1.5:2 were used as precursors. The precursors were ground using high-energy ball milling at 400 rpm for 12 h. The mixture was calcined at 300 °C for 6 h under Ar flow. After calcination, the powder was well-ground and high-energy ball milled with PA. The resulting powder was pelletized under a pressure of 200 kg cm^{-2} and sintered at 550 °C for 12 h under Ar flow. The carbon-coated sample included approximately 6 wt % carbon.

Materials Characterization. The $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ was analyzed using XRD (PANalytical, Empyrean) with Cu K α radiation ($\lambda = 1.54178 \text{ \AA}$) in the 2θ range of 8–80° with a step size of 0.13°. Rietveld refinement of the XRD data was performed using the FullProf software. The valence states of Mn and Co in the structure of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ were determined from X-ray absorption near-edge structure (XANES) analysis performed on beamline 6D at the Pohang Accelerator Laboratory, South Korea. The Mn K-edge and Co K-edge edge spectra were collected in the transmission mode with an electron energy of 2.5 GeV and a current of 200 mA. Mn and Co reference spectra were obtained simultaneously from Mn and Co metal foil, respectively. Particle distribution and particle size measurements were performed using

scanning electron microscopy (SEM) (Hitachi S-4700) and high-resolution transmission electron microscopy (TEM). The atomic ratios of elements such as Na, Mn, and Co were determined using inductively coupled plasma optical emission spectrometer at Daeduck Analytical Research Institute. The Operando XRD patterns of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ were analyzed using XRD (PANalytical, Empyrean) with Cu K α radiation ($\lambda = 1.54178 \text{ \AA}$) in the 2θ range of 18–38° with a step size of 0.026°. During operando XRD experiments, we used the Operando battery cell (ISBC) for XRD made by PANalytical with PDC TECH. To protect the cell system from water, we used a beryllium (Be) window.

Electrochemical Characterization. The active materials, Super P carbon black, and polyvinylidene fluoride were blended in a weight ratio of 8:1:1 in *N*-methyl-2-pyrrolidone. The slurry was applied on Al foil and subsequently dried at 100 °C for 24 h under vacuum. After drying, the electrode was punched into discs (10ϕ , 3 mg). Electrochemical tests were performed on R2032-type coin cells assembled in an Ar-filled glove box. The electrolyte used was 0.5 M NaPF_6 in polypropylene carbonate with 2 vol % fluoroethylene carbonate. The fabricated cathode was separated from a Na metal anode by a Whatman GF/F glass fiber separator.

Computational Details. All density functional theory (DFT) calculations were performed using the Vienna Ab initio Simulation Package (VASP).⁴⁸ We used projector-augmented wave pseudopotentials⁴⁹ with a plane-wave basis set as implemented in VASP. Perdew–Burke–Ernzerhof parametrization of the generalized gradient approximation (GGA)⁵⁰ was used for the exchange–correlation functional. For DFT calculations, a $4 \times 3 \times 2$ *k*-point grid was used to calculate a $1 \times 1 \times 1$ supercell structure of $\text{Na}_{16-x}[\text{Mn}_6\text{Co}_6](\text{P}_4\text{O}_{16})_2(\text{P}_8\text{O}_{28})$ that is the four formula units of $\text{Na}_{4-x}[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$. Using the cluster-assisted statistical mechanics (CASM) software, we can make various Na/vacancy configurations of each $\text{Na}_{16-x}[\text{Mn}_6\text{Co}_6](\text{P}_4\text{O}_{16})_2(\text{P}_8\text{O}_{28})$ composition. Among these numerous Na/vacancy configurations on each composition, the 30 lowest electrostatic energy configurations were confirmed for each composition, as calculated using the Ewald summation method,⁵¹ and then, we calculated their formation energies using first-principles calculations. Because almost all blue points are thermodynamically stable, their formation energies are negative values. In case of the very unstable configurations, their formation energies are positive values. Because difference of Na/vacancy configuration results in small difference of formation energies despite the same atomic composition, one particular Na/vacancy configuration on $\text{Na}_{16-x}[\text{Mn}_6\text{Co}_6](\text{P}_4\text{O}_{16})_2(\text{P}_8\text{O}_{28})$ composition is relatively more stable than the other configurations. Thus, we can determine the most stable configuration on $\text{Na}_{4-x}[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ compositions ($0 \leq x \leq 3$). The GGA + *U* method⁵² was adopted to address the localization of the d-orbital in Mn and Co ions, with U_{eff} value of 4.5 and 5.7 eV, determined in the previous studies.^{34,53} An appropriate number of *k*-points and a kinetic energy cutoff of 500 eV are used in all calculations. All structures are optimized until the force in the unit cell converges within $0.03 \text{ eV \AA}^{-1}$.

Nudged elastic band (NEB) calculations⁵⁴ were performed to determine the activation barrier for Na diffusion in the $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ structure. To perform the calculations, five intermediate images are generated between each Na site. These structures are then calculated by the NEB algorithm with the fixed lattice parameters and free internal atomic positions.

■ RESULTS AND DISCUSSION

$\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ was successfully synthesized via a simple solid-state reaction (see the [Experimental section](#)). The morphology and atomic composition of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ were investigated using TEM combined with energy-dispersive X-ray spectroscopy (EDS) and inductively coupled plasma–atomic emission spectroscopy. The TEM images presented in [Figures 1a](#) and [S1](#) show the uniform distribution of Na, Mn, and Co elements in a

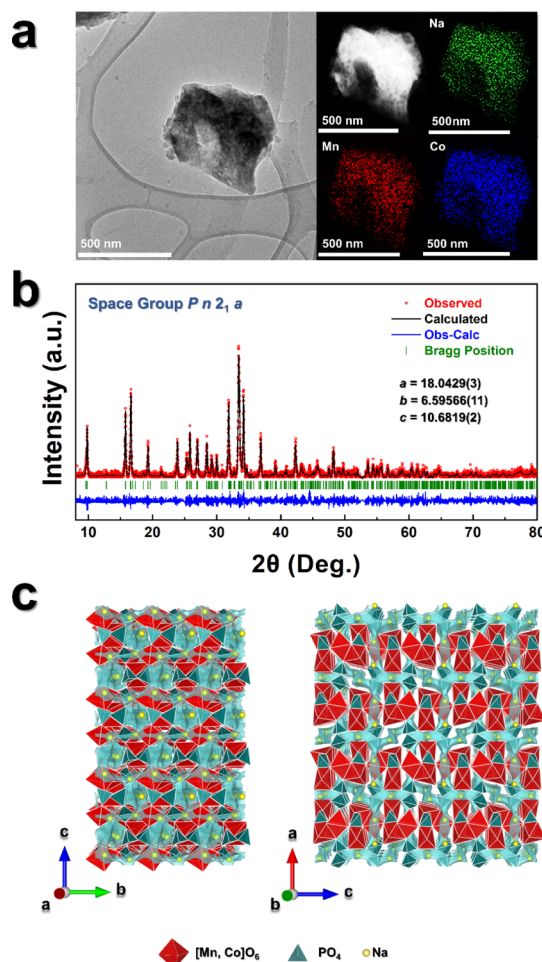


Figure 1. (a) TEM image and EDS elemental mappings of $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$. (b) Rietveld refinement of XRD pattern of $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$. (c) BVEL analysis of $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ with all possible Na-ion sites in the crystal structure.

$Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ particle of submicron size. The Na/Mn/Co atomic ratio was verified to be 3.98:1.49:1.50 by the chemical analysis (Table S1). To obtain structural information for $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$, Rietveld refinement of the XRD data was performed using the determined chemical composition. Structural information of $Na_4Mn_3(PO_4)_2(P_2O_7)$ was used, and the Rietveld refinement was carried out by placing cobalt (Co) at the manganese (Mn) 4a sites and setting each occupancy to 0.5. The refined results of $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ indicated that the compound crystallized in a pure orthorhombic structure with space group $Pn2_1a$ (Figure 1b) with acceptable reliable factors ($R_p = 5.91\%$, $R_1 = 5.51\%$, $R_F = 5.17\%$, and $\chi^2 = 6.71\%$). The calculated lattice parameters of $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ were $a = 18.0429(3)$ Å, $b = 6.59566(11)$ Å, and $c = 10.6819(2)$ Å, which are smaller than those of $Na_4Mn_3(PO_4)_2(P_2O_7)$ but larger than those of $Na_4Co_3(PO_4)_2(P_2O_7)$ because of the difference in the ionic radii of Mn^{2+} (~ 0.83 Å) and Co^{2+} (~ 0.745 Å). The detailed structural information for $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ is provided in Table 1. For verification of the structural information of $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ using Rietveld refinement, we used the structural information of $Na_4M_3(PO_4)_2(P_2O_7)$ ($M = Fe, Mn, Co, Ni, Mg$), and it

was reported that $Na_4M_3(PO_4)_2(P_2O_7)$ is a mixed-polyanion or mixed-anion phosphate.^{21,23} As presented in Figure 1b and Table S2, the structural information of $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ is well consistent with that of previously reported $Na_4M_3(PO_4)_2(P_2O_7)$, which indicates that $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ can be identified as one of the mixed-polyanions. In addition, bond-valence energy landscape (BVEL) analysis provided additional information on possible Na^+ ionic diffusion paths in $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$. As observed in Figure 1c, large three-dimensional pathways for Na^+ diffusion are present in the structure, which can transport Na^+ during the electrochemical reaction of $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ in Na cells.

To predict the properties and reaction mechanism of $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ during Na^+ de/intercalation, first-principles calculation was performed using the structural information obtained from the Rietveld refinement and XRD analysis. We assumed several Na^+ vacancy configurations for $Na_{4-x}[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ ($0 \leq x \leq 3$) using CASM software and then calculated and arranged the formation energies of each configuration. The theoretical redox potential of $Na_{4-x}[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ was calculated using the following equation

$$V = -\frac{(E[Na_{x_2}[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)] - E[Na_{x_1}[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)] - (x_2 - x_1)E(Na))}{((x_2 - x_1)F)} \quad (1)$$

where V is the average redox potential of $Na_x[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$, $E[Na_x[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)]$ is the formation energy of the most stable configuration of $Na_x[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ calculated using first-principles calculation, $E(Na)$ is the formation energy of Na metal, and F is the Faraday constant. Figure 2a presents the convex-hull plot of the formation energies of $Na_{4-x}[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ as a function of the Na content. We predicted the existence of several intermittent phases between $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ and $Na_1[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$, which implies a solid-solution (one-phase) reaction of $Na_{4-x}[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ during Na^+ de/intercalation. As observed in Figure 2b, the charge/discharge profile of $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ obtained at C/10 ($1\text{ C} = \sim 128\text{ mA g}^{-1}$) was consistent with the theoretical redox potential predicted by first-principles calculation. In addition, it was predicted through first-principles calculations that $\sim 4.54\text{ V}$ (vs Na^+/Na) is required for further Na deintercalation from $Na_{1.25}[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$. However, it was reported that the side reactions such as electrolyte-decomposition could occur at the voltage range above 4.5 V (vs Na^+/Na).^{55,56} Thus, it is reasonable that approximately 2.7 mol Na^+ ions per formula unit (116 mA h g^{-1}) are involved in the reversible de/intercalation process for $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ with an average operation voltage of $\sim 4.0\text{ V}$ versus Na^+/Na , which is higher than that of $Na_4Mn_3(PO_4)_2(P_2O_7)$ ($\sim 3.8\text{ V}$ vs Na^+/Na). In addition, the smooth feature of the voltage profile and two voltage plateaus indicate that the Na^+ de/intercalation process is associated with a solid-solution (one-phase) reaction.

Electrochemical tests on $Na_4[Mn_{1.5}Co_{1.5}](PO_4)_2(P_2O_7)$ were performed using a R2032-type coin cell using Na metal as the counter electrode. The details of the cell configuration are provided in the Experimental section. Figure

Table 1. Detailed Structural Information of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ through Rietveld Refinement^a

atom	Wyckoff position	x	y	z	B_{iso}	occupancy
Mn1	4a	0.3400(4)	0.099(2)	0.5089(11)	1.4(3)	0.5
Mn2	4a	0.1356(3)	0.589(3)	0.4890(11)	0.28(6)	0.5
Mn3	4a	0.2412(5)	0.3190(19)	0.7492(18)	0.55(6)	0.5
Co1	4a	0.3400(4)	0.099(2)	0.5089(11)	1.4(3)	0.5
Co2	4a	0.1356(3)	0.589(3)	0.4890(11)	0.28(6)	0.5
Co3	4a	0.2412(5)	0.3190(19)	0.7492(18)	0.55(16)	0.5
P1	4a	0.3044(6)	0.580(4)	0.515(2)	0.31(6)	1
P2	4a	0.1770(6)	0.078(4)	0.4852(18)	0.31(6)	1
P3	4a	0.5667(6)	0.4633(17)	0.745(2)	0.31(6)	1
P4	4a	0.4476(7)	0.133(2)	0.7339(16)	0.31(6)	1
O1	4a	0.2310(16)	0.567(6)	0.574(2)	0.50(2)	1
O2	4a	0.3482(15)	0.462(4)	0.451(3)	0.50(2)	1
O3	4a	0.3474(15)	0.773(4)	0.543(3)	0.50(2)	1
O4	4a	0.2373(16)	0.608(6)	0.385(2)	0.50(2)	1
O5	4a	0.2215(16)	0.083(6)	0.583(2)	0.50(2)	1
O6	4a	0.1338(14)	−0.079(4)	0.501(3)	0.50(2)	1
O7	4a	0.2391(16)	0.044(5)	0.375(2)	0.50(2)	1
O8	4a	0.1325(14)	0.280(4)	0.465(3)	0.50(2)	1
O9	4a	0.4812(15)	0.359(5)	0.713(3)	0.50(2)	1
O10	4a	0.5439(16)	0.571(5)	0.871(2)	0.50(2)	1
O11	4a	0.6146(12)	0.295(4)	0.744(4)	0.50(2)	1
O12	4a	0.5808(16)	0.625(5)	0.632(2)	0.50(2)	1
O13	4a	0.4564(17)	0.123(6)	0.864(2)	0.50(2)	1
O14	4a	0.3706(12)	0.173(4)	0.700(3)	0.50(2)	1
O15	4a	0.4816(17)	0.021(4)	0.639(2)	0.50(2)	1
Na1	4a	0.4932(16)	0.798(5)	0.980(2)	0.40(3)	1
Na2	4a	0.2925(9)	0.841(3)	0.757(3)	0.40(3)	1
Na3	4a	0.4049(9)	0.436(3)	0.258(3)	0.40(3)	1
Na4	4a	0.4637(11)	0.692(3)	0.533(2)	0.40(3)	1

^aCrystal structure: orthorhombic. Space group: $Pn2_1a$ (33). Lattice parameters: $a = 18.0429(3)$, $b = 6.59566(11)$, $c = 10.6819(2)$

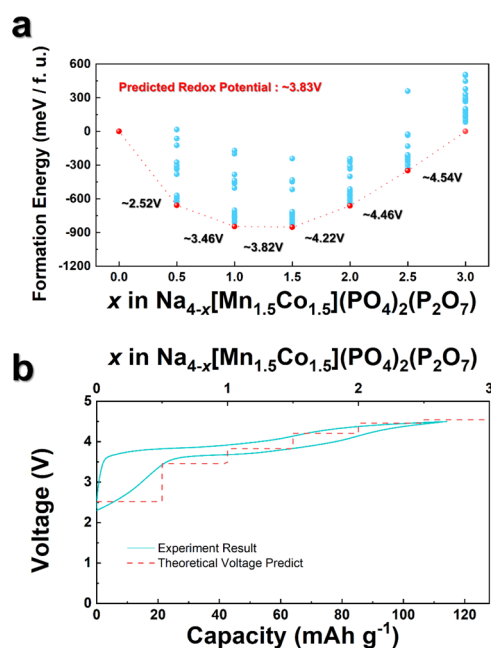


Figure 2. (a) Formation energy and derived theoretical voltage of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$. (b) Comparison of theoretical voltage predictions and experimental results for $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$.

3a presents the charge/discharge profiles and resulting capacities of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ measured at

various C-rates between 2.3 and 4.5 V (1 C = 128 mA h g^{-1}). $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ exhibited reasonable rate performance at various C-rates (116 mA h g^{-1} at 1 C, 97 mA h g^{-1} at 5 C, and 84 mA h g^{-1} at 10 C, with the capacity at 10 C corresponding to 73% of that at C/10). These capacities were maintained for five cycles at each C-rate (Figure 3b). Notably, the capacity obtained at 0.1 C (116 mA h g^{-1}) is the highest among $\text{Na}_4\text{M}(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (M: Mn, Co) compounds. To confirm the excellent power-capability of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$, first-principles calculation was employed to theoretically predict Na^+ diffusion in the structure using the NEB method (Figure 3c,d). Unexpectedly, the required energy was only ~ 196 meV for Na^+ diffusion along the $\text{Na}_{1-1}-\text{Na}_{2-1}$ pathway in the $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ structure (Figure 3c), resulting in facile diffusion of Na^+ into the host structure and leading to the high rate performance of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$. The high-rate cycling at 2 C elucidates the validity of the low-energy diffusion barrier for $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$, which exhibited outstanding capacity retention of approximately 75% for 300 cycles (Figure 3e) better than the cyclability of $\text{Na}_4\text{Mn}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ tested under the same condition (Figure S2). To the best of our knowledge, this is the first report to demonstrate high performance combined with high operation voltage for $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ compounds.

The outstanding electrode performance of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ was speculated to result from a small structural change during Na^+ de/intercalation; hence, operando XRD

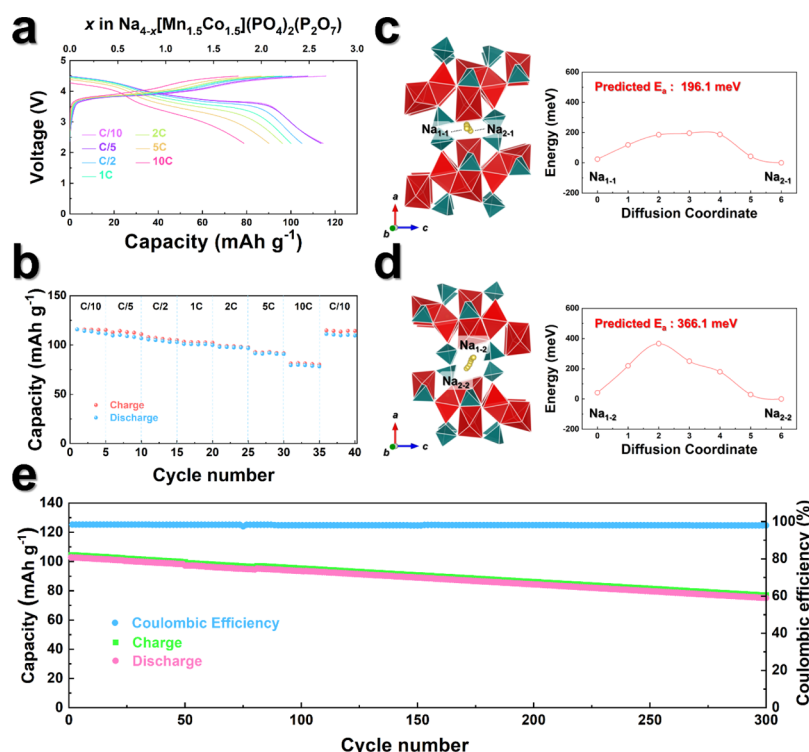


Figure 3. (a) Charge/discharge profiles of Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) in the range of 2.3–4.5 V at various C-rates. (b) Power capability of Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇). (c,d) Diffusion pathway with energy landscape determined using the NEB method for Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) structure. (e) Cycle performance of Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) at 2 C over 300 cycles.

analysis of Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) during the first cycle was performed (Figure 4a). Although no new peaks were observed to evolve, the original peaks gradually shifted toward higher and lower angle during charge and discharge, respectively. These changes are typically observed for a one-phase (solid-solution) reaction, namely, the solid-solution reaction of Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) (0 ≤ x ≤ 2.7), as predicted from the first-principles calculation (Figure 2a,b). In addition, merging and splitting of the (004) and (222) peaks at 33.5° (2θ) was observed. In the range of x = 0–1.5 for Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇), merging of the peaks at the lower voltage plateau occurred on charge, whereas splitting of the peak into (004) and (222) peaks was observed in the range of x = 1.5–2.7. This behavior was reversible on discharge. The calculated lattice parameters varied monotonously on charge analyses, which is typically observed for a one-phase (solid-solution) reaction (Figure 4b). Moreover, as presented in Figure S3, we performed the Rietveld refinement on the full charged Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇), which indicates that despite splitting of the (004) and (222) peaks, overall crystal structure of Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) was well retained after full charging to 4.5 V (vs Na⁺/Na) without formation of new phases. As Na⁺ ions were extracted from the crystal structure, the c lattice parameter linearly increased, while the a and b lattice parameters decreased. The total volume change was calculated to be 5.7%, which is far smaller than that of other Na₄M(PO₄)₂(P₂O₇) (M: Mn, Co) compounds and minimized the strain generated by Na⁺ insertion into/extraction out of the host structure. The low energy barrier and small volume change of the structure may have contributed to the excellent long-term electrode performance at high rates.

The main redox center of Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) was investigated using XANES spectroscopy during charging (Figure 5). The Mn K-edge gradually shifted to higher energy on charging to 4.2 V [Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) (x = 0–1.5)] (Figure 5a), whereas there was almost no movement of the Co K-edge in this operation range (Figure 5b). In contrast, the Co K-edge actively shifted to higher energy (Figure 5b), whereas negligible change was observed for the Mn–K edge when charging to 4.5 V [Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) (x = 1.5–2.7)]. These results indicate that during 2.7 mol Na⁺ deintercalation from Na₄[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇), the Mn²⁺/Mn³⁺ redox reaction first occurs on the lower voltage plateau followed by the Co²⁺/Co³⁺ redox reaction on the upper voltage plateau.

In addition, the activities of both the Mn²⁺/Mn³⁺ and Co²⁺/Co³⁺ redox pairs in Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) during charge/discharge were confirmed through comparison of the integrated spin moments of Mn and Co ions between Na₄[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) and Na₁[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) using first-principles calculation. As shown in Figure 6, the total electron spin counts of Mn²⁺ and Co²⁺ ions on Na₄[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) were confirmed to be five and three, respectively. During Na⁺ deintercalation, the total electron spin count of the Mn ions decreased to four from five and that of the Co ions increased to four from three. This behavior clearly demonstrates that during the charging/discharging process, Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) is activated by the Mn²⁺/Mn³⁺ and Co²⁺/Co³⁺ redox reactions, which is consistent with the XANES data presented in Figure 5.

As observed in Figure 4a, the solid-solution reaction of Na_{4-x}[Mn_{1.5}Co_{1.5}](PO₄)₂(P₂O₇) differs from that of Na₄Mn₃(PO₄)₂(P₂O₇), which undergoes a two-phase reaction.

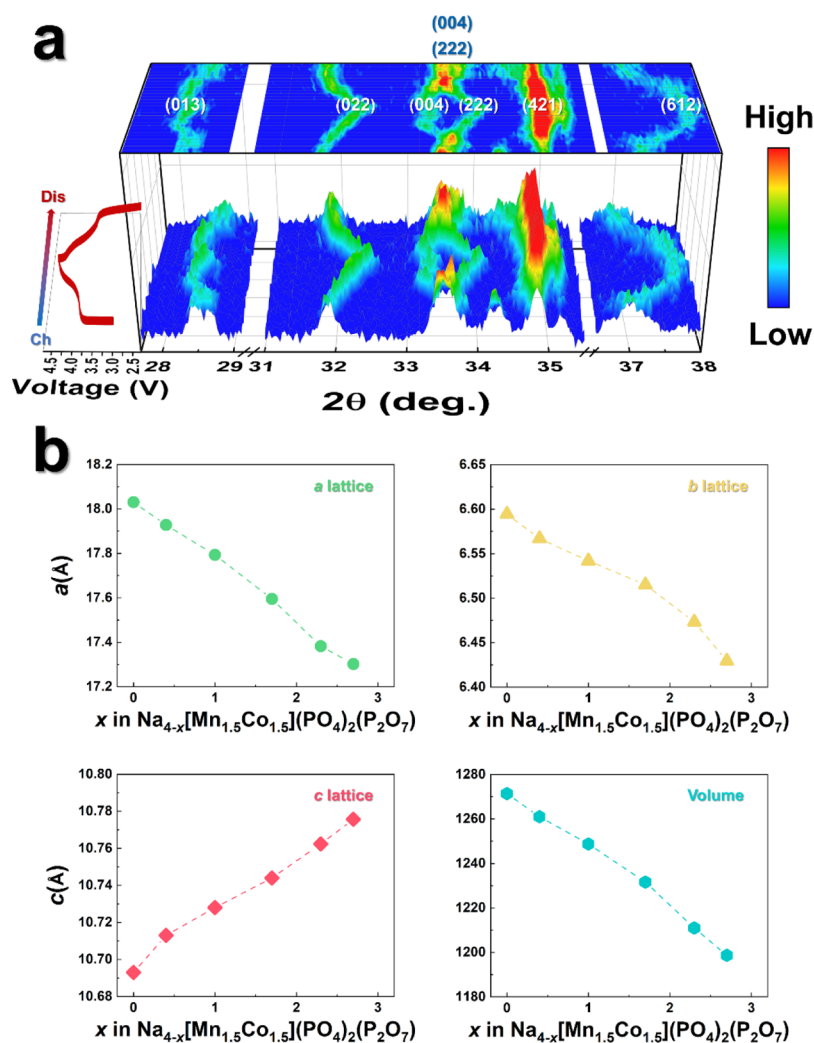


Figure 4. (a) Operando XRD patterns collected during charge/discharge of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ at a current rate of $C/10$ in the voltage range between 2.3 and 4.5 V. (b) Changes of lattice parameters a , b , c and volume of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a function of Na content.

Indeed, both compounds possess Mn^{3+} ions in the charged state, whereas Mn^{3+} is present together with Co^{3+} for $\text{Na}_{1.3}[\text{Mn}^{3+}_{1.5}\text{Co}^{3+}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$. For Mn^{3+} ions, Jahn–Teller distortion is known to result in elongation of the Mn–O distance along the z -axis in the MnO_6 octahedron. It has been reported that the large/anisotropic structural change and long-range ordering between Na ions and vacancies in the structure might result from Jahn–Teller distortion of Mn^{3+} . This finding implies that a certain Na^+ /vacancy configuration in the crystal structure of Mn-based electrode materials may be required to reduce the structural instability arising from Jahn–Teller distortion of Mn^{3+} . In this study, the presence of Co^{3+} reduces the concentration of Mn^{3+} in the desodiated state as they share the oxygen in their octahedral environment, such that the Jahn–Teller effect may be mitigated in the structure, enabling long-term cycling at 2 C without changing the structure. The morphology and crystal structure of the cycled $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ were analyzed using SEM, TEM, and Rietveld refinement of XRD data (Figures 7, S4, and Tables S2 and S3). Compared with the as-synthesized electrode, notable degradation of the structure and morphology of the post-cycled electrode was not observed, which

further demonstrates the outstanding cycle performance of $\text{Na}_4\text{Mn}_{1.5}\text{Co}_{1.5}(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ in Na cells.

CONCLUSIONS

In summary, we investigated the structural and electrochemical properties of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ for potential use as a high-energy-density cathode material for NIBs. Based on the structural information determined from Rietveld refinement of XRD data, we predicted various theoretical properties of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ using first-principles calculation. It was verified that at a current rate of $C/10$ ($1\text{ C} = 128\text{ mA g}^{-1}$), $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ delivered a specific capacity of $\sim 116\text{ mA h g}^{-1}$, corresponding to $\sim 2.7\text{ mol Na de/intercalation}$, with a high average operation voltage of $\sim 4.0\text{ V}$ (vs Na^+/Na) resulting from the $\text{Mn}^{2+}/\text{Mn}^{3+}$ and $\text{Co}^{2+}/\text{Co}^{3+}$ redox reactions. Even at 10 C, the specific capacity of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ was still maintained up to $\sim 73\%$ of the capacity measured at $C/10$. This outstanding power-capability was clearly confirmed by NEB analyses with first-principles calculation predicting low activation barrier energies for Na^+ diffusion in the $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ structure. Furthermore, over 300 cycles at 2 C, $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ exhibited

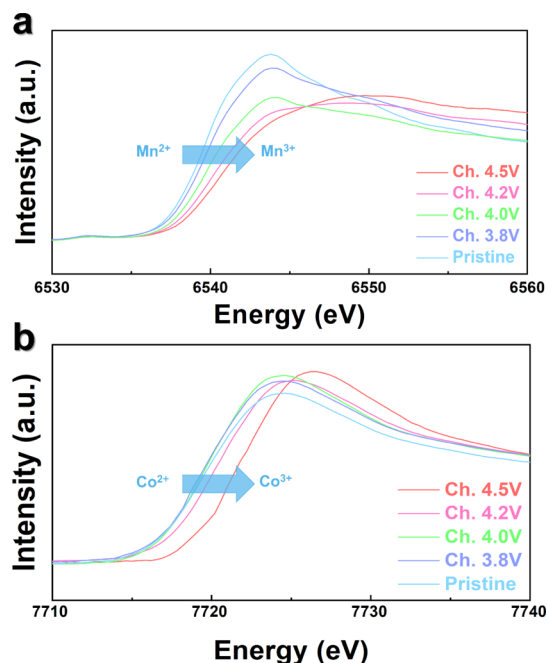


Figure 5. XANES analyses of (a) Mn and (b) Co K-edge for $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$.

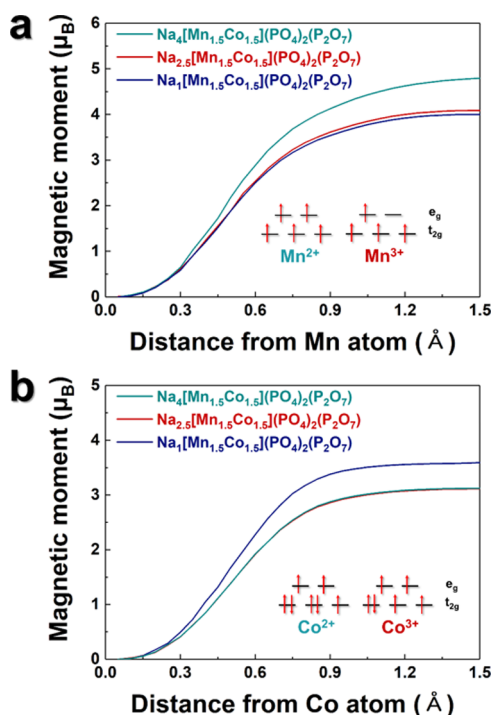


Figure 6. Spin integration of (a) Mn ions and (b) Co ions in $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$, $\text{Na}_{2.5}[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$, and $\text{Na}_1[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$.

capacity retention of $\sim 75\%$ compared with the initial capacity with a high Coulombic efficiency of greater than $\sim 99\%$. We verified that the small volume change during charge/discharge and suppression of the Jahn–Teller distortion by Mn^{3+} ions may enable improved cyclability of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ relative to that of $\text{Na}_4\text{Mn}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$. We believe that our findings will provide insight for the

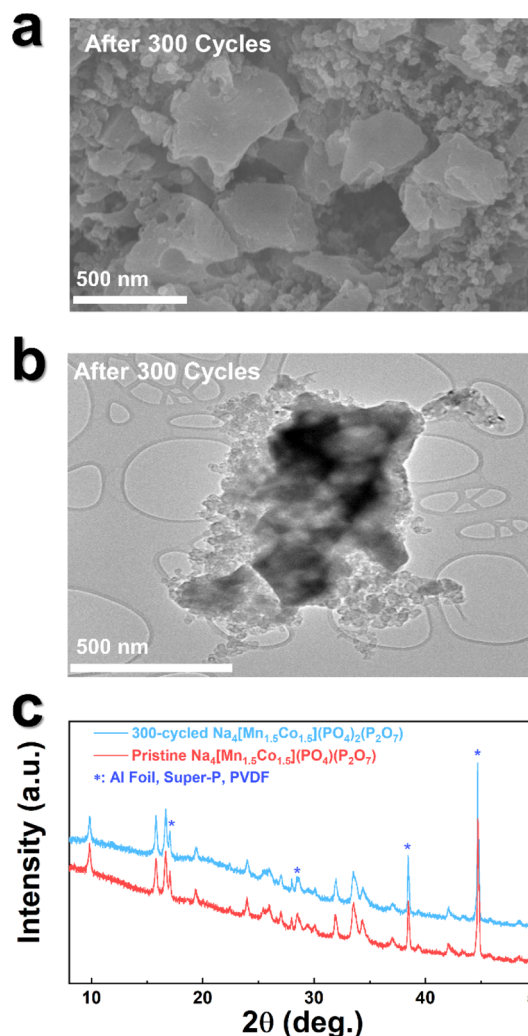


Figure 7. (a) SEM image, (b) TEM image, and (c) XRD patterns of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ electrode after cycling for 300 cycles.

development of novel cathode materials for NIBs as well as for various other NIB-related studies.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.9b04944>.

SEM image of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$; cycling performance of the $\text{Na}_4\text{Mn}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ at 2 C over 200 cycles; Rietveld refinement of XRD pattern of the full charged $\text{Na}_{4-x}[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (to 4.5 V vs Na^+/Na) ($R_p = 1.25\%$, $R_1 = 0.044\%$, $R_F = 0.0649\%$, $\chi^2 = 1.60\%$); Rietveld refinement of XRD pattern of 300 cycled $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ ($R_p = 1.34\%$, $R_1 = 1.56\%$, $R_F = 1.34\%$, $\chi^2 = 1.71\%$); ICP analyses on the atomic ratio of Na, Mn, and Co in the pristine sample of $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$; detailed structural information of 300 cycled $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ electrodes; and Rietveld refinement results of Pristine and 300 cycled $\text{Na}_4[\text{Mn}_{1.5}\text{Co}_{1.5}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (PDF)

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Notes

The authors declare no competing financial interest.

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